Network Communication

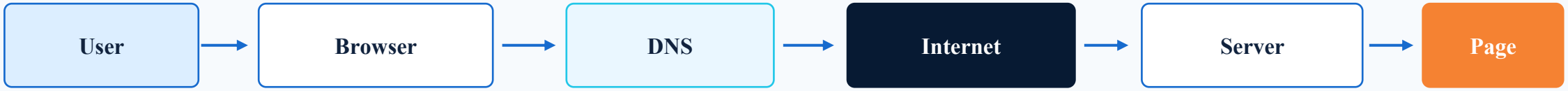
How Data Moves Across Networks



DSI102 Introduction to Digital Technology
Dr. Nantapong Keandoungchun

Opening Scenario

What actually happens when a user opens a website?



The lecture is rearranged around one story: an application request becomes packets, travels across networks, and returns as a response.

Learning Outcomes

After this lecture, students should be able to explain network communication, not just name protocols.

Layered models

Explain OSI and TCP/IP responsibilities

Addressing

Interpret MAC, IP, port, CIDR and subnet mask basics

Packets

Describe headers, payloads, frames, segments and bits

Transport

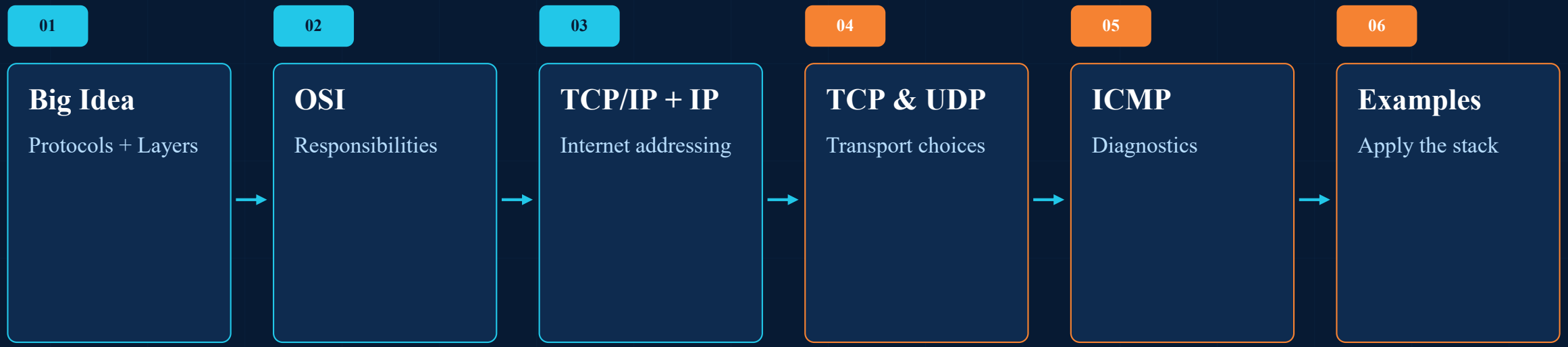
Compare TCP and UDP with examples

Diagnostics

Use ICMP conceptually through ping and traceroute

Rearranged Lecture Roadmap

From big idea to layers, then protocols, then diagnostics



SECTION 1

Big Idea

Protocols, layers and encapsulation before detailed models



Connected ≠ Communicating

A physical link does not guarantee mutual understanding

Connection

Devices can send signals through a medium.

Examples:
- cable
- Wi-Fi
- fiber
- radio link

Communication

Devices follow rules so that data has meaning.

Examples:
- format
- timing
- addressing
- error handling

A network needs both infrastructure and agreed communication rules.

Protocol = Rulebook

Protocols coordinate format, timing, addressing and interpretation

Syntax

What does the message look like?

Semantics

What does each field mean?

Timing

When should messages be sent or answered?

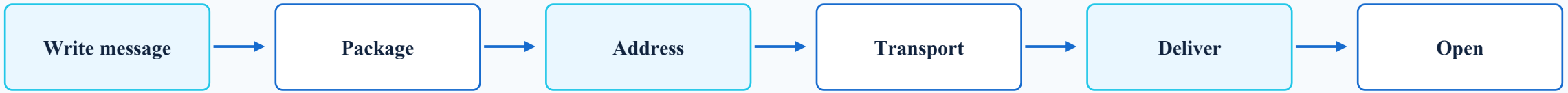
Action

What should the receiver do next?

Examples: HTTP defines web requests; TCP defines connection behavior; IP defines packet delivery.

Why Do Networks Use Layers?

Large communication problems become smaller responsibilities



Modularity

Each layer solves one type of problem.

Standardization

Different vendors can interoperate.

Troubleshooting

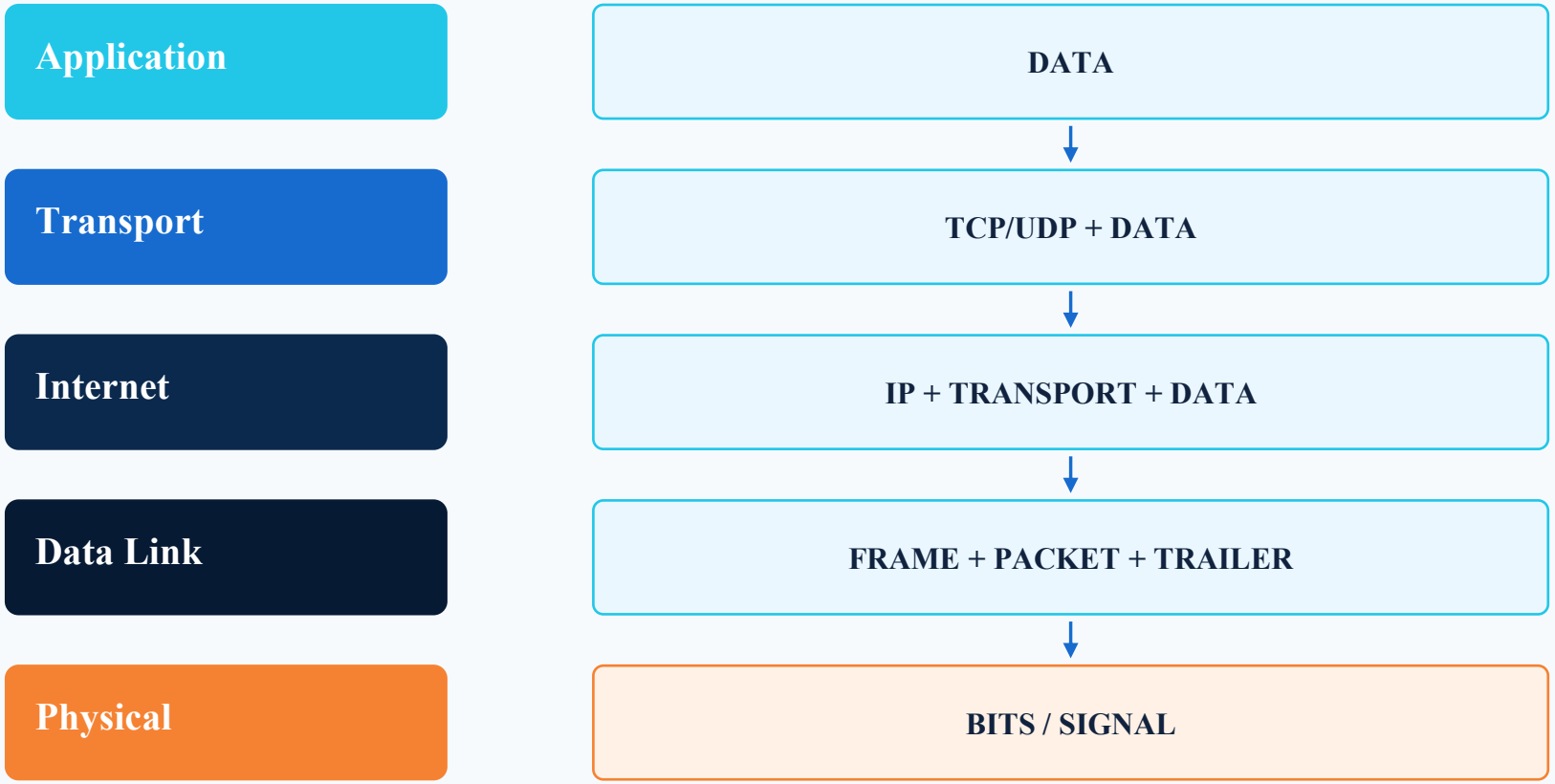
Problems can be isolated by layer.

Evolution

One layer can improve without replacing everything.

Encapsulation: Data Gets Wrapped

Each layer adds information needed for communication



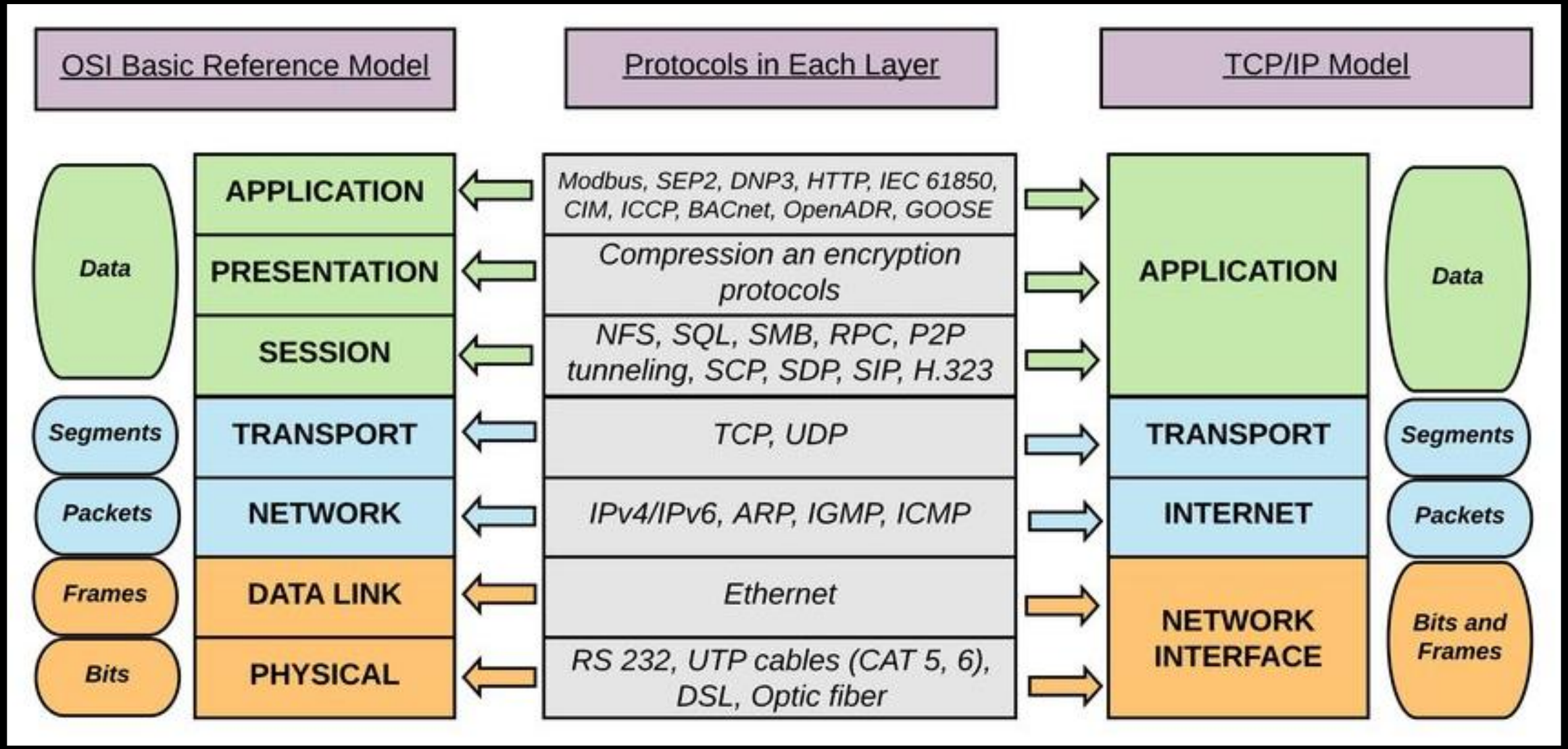
Encapsulation explains why the same data may be called data, segment, packet, frame, or bits at different layers.

Vocabulary: Data Units by Layer

Packet, frame, segment and bits are not the same thing

Layer group	Common data unit	Meaning
Application / Presentation / Session	Data	Information used by applications
Transport	Segment / Datagram	Transport-layer unit
Network / Internet	Packet	Routed using IP information
Data Link	Frame	Local link delivery unit
Physical	Bits	Signals on the medium

Misconception to avoid: a packet is not automatically the same thing as a frame.



Communication protocols

https://www.researchgate.net/figure/The-logical-mapping-between-OSI-basic-reference-model-and-the-TCP-IP-stack_fig2_327483011

SECTION 2

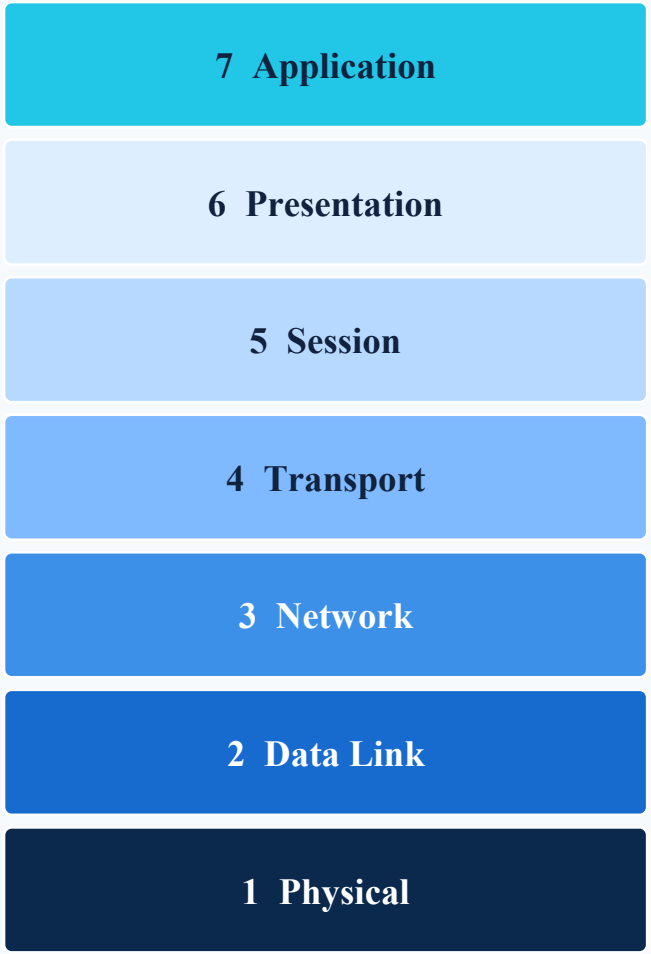
OSI Model

Communication responsibilities from application to physical signal



OSI Model Overview

Seven conceptual layers organize network communication



User network services

Data representation

Session control

End-to-end transport

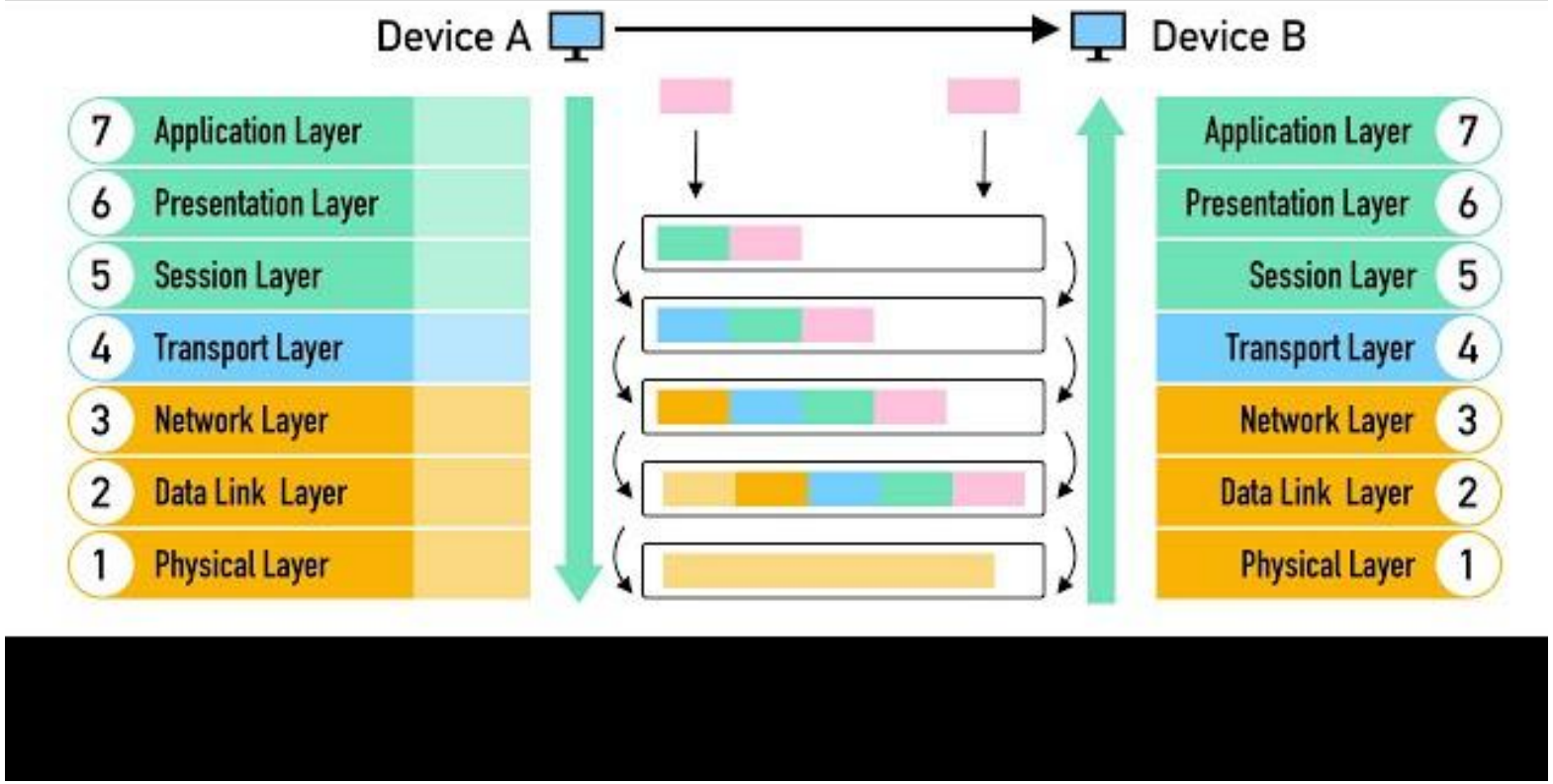
Routing and addressing

Local frame delivery

Signal transmission

OSI helps us think: each layer has a responsibility and communicates with neighboring layers.

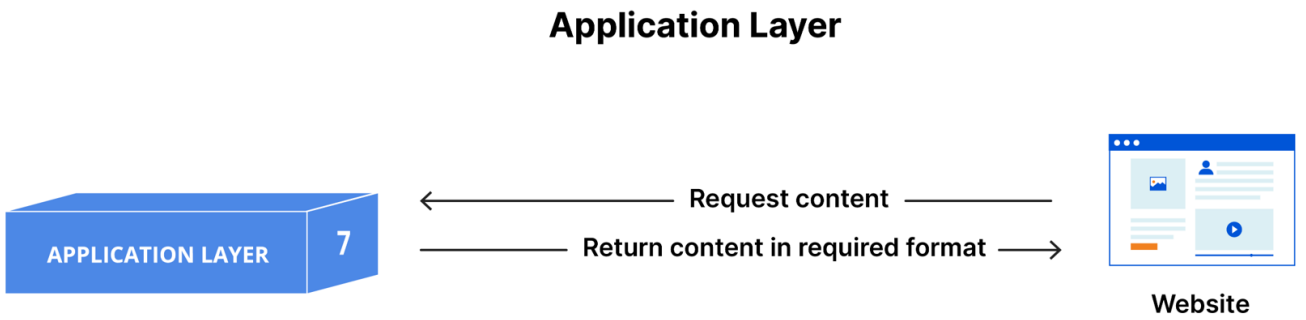
What is OSI Model?



https://www.youtube.com/watch?v=0y6FtKsg6J4

Application Layer

Where user-facing software starts network communication



Web Browser

HTTP / HTTPS request

Email Client

SMTP / IMAP / POP concept

File Transfer

FTP / SFTP concept

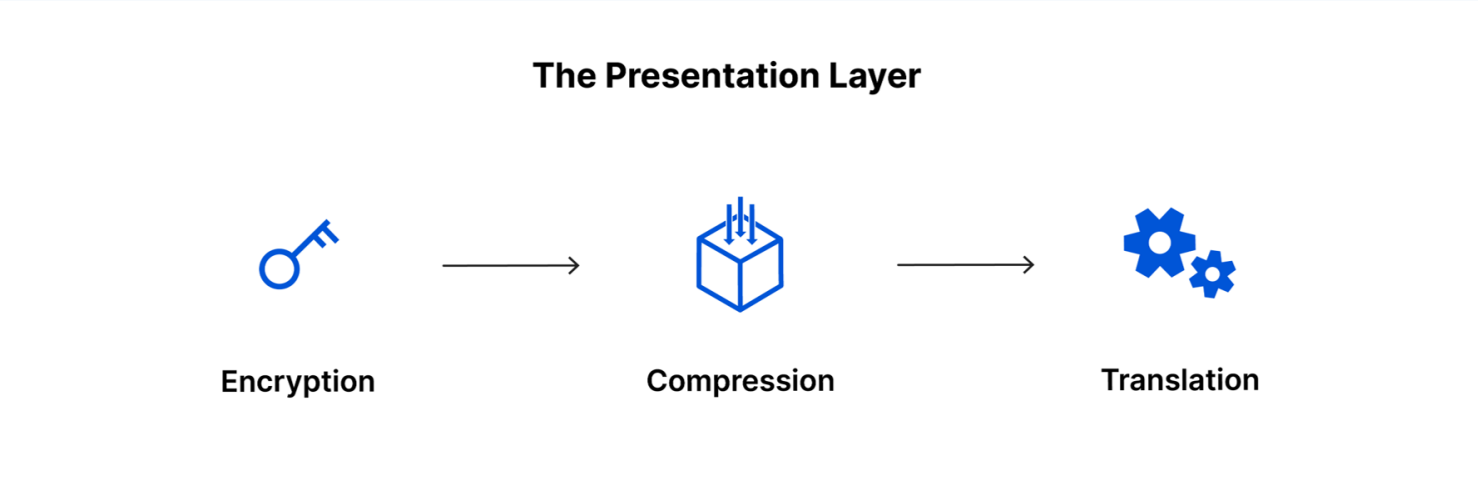
DNS Client

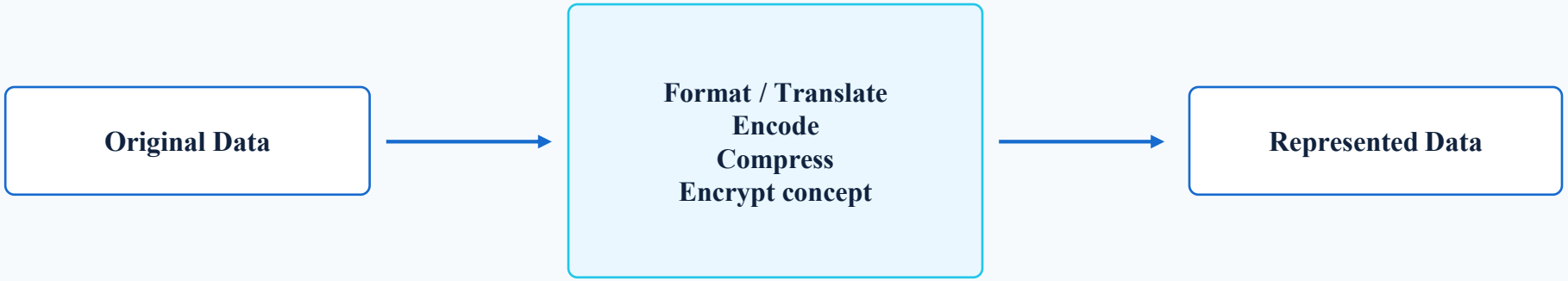
Asks for a name-to-address lookup

The application layer is not the app itself; it is the network service interface used by software.

Presentation Layer

Preparing data representation for the receiving system





For this course, treat the presentation layer as the place where data representation is prepared and interpreted.

Session Layer and Duplex Modes

Managing ongoing conversations between related streams

Session management

Establish
Maintain
Checkpoint
Suspend / restart
Terminate communication sessions

Simplex

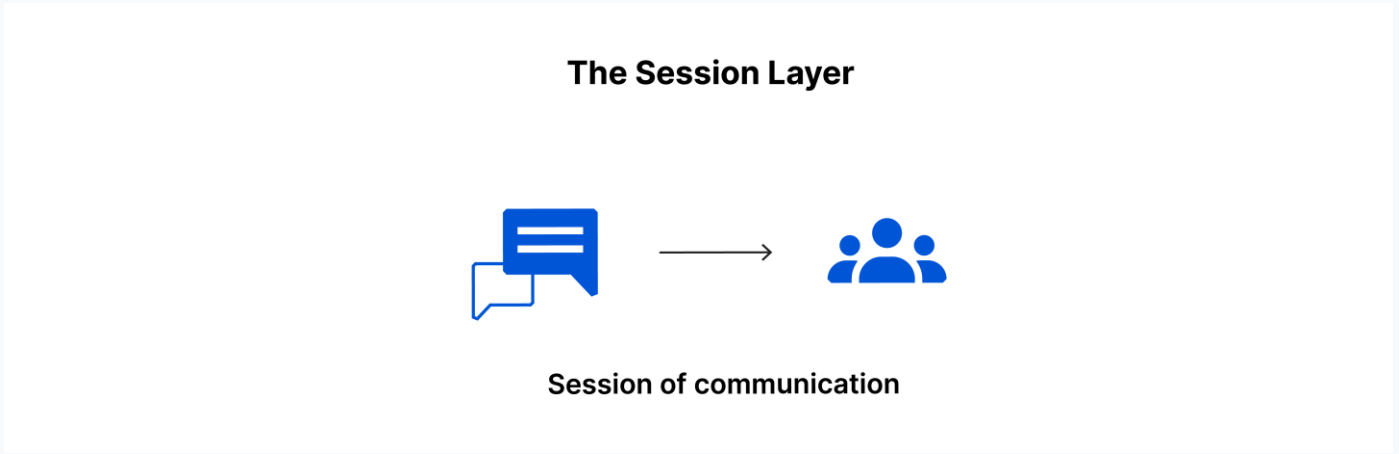
A → B

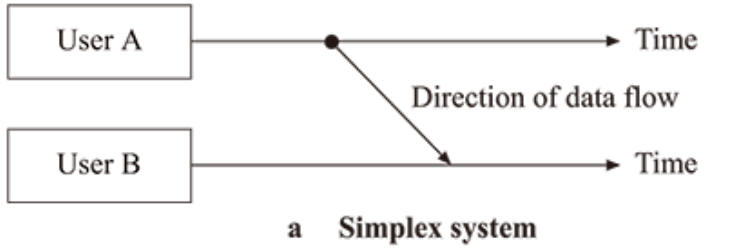
Half-Duplex

A ↔ B
one direction at a time

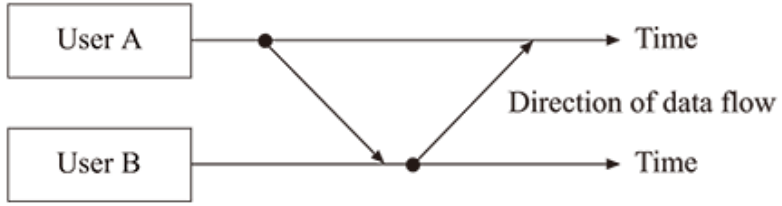
Full-Duplex

A ⇄ B
simultaneously

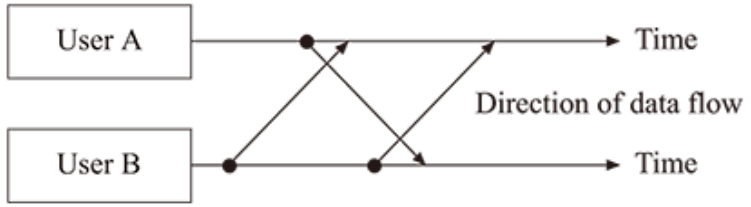




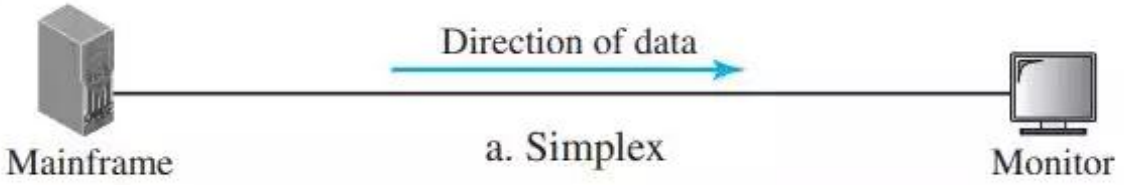
a Simplex system

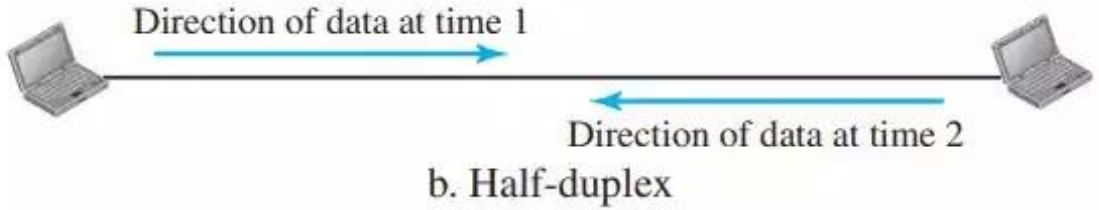


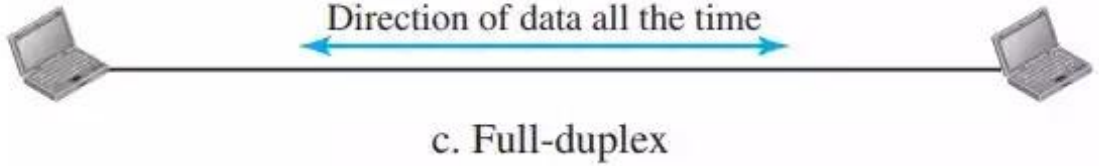
b Half-duplex system



c Full-duplex system



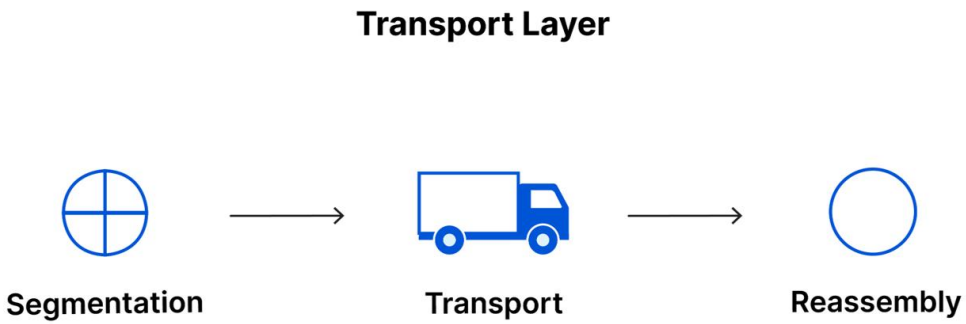




Simplex vs Half-Duplex vs Full-Duplex

Transport Layer

End-to-end communication between application processes



Segmentation

Break large data into smaller pieces

Acknowledgement

Confirm received data when required

Multiplexing

Support multiple application conversations

Ports

Identify the application process

The transport layer answers: which application process should receive this data, and how should delivery behave?

COTP and OSI Transport Classes

A concise view of connection-oriented transport in the OSI family

COTP

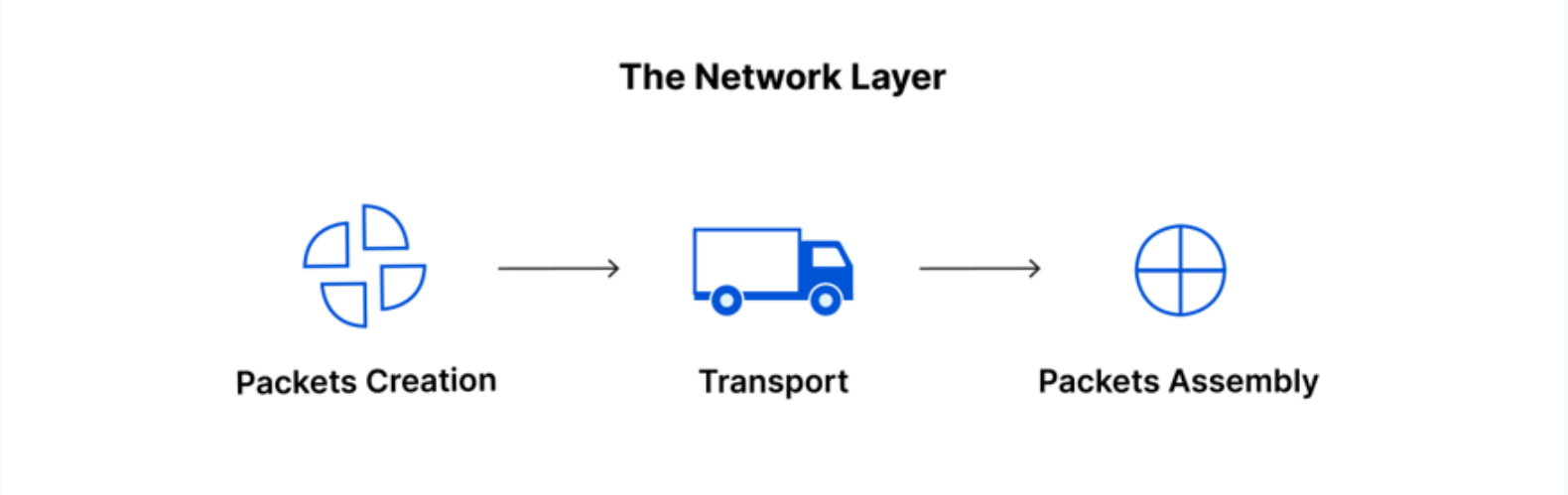
Connection-Oriented Transport Protocol. Included because it appears in the source lecture.

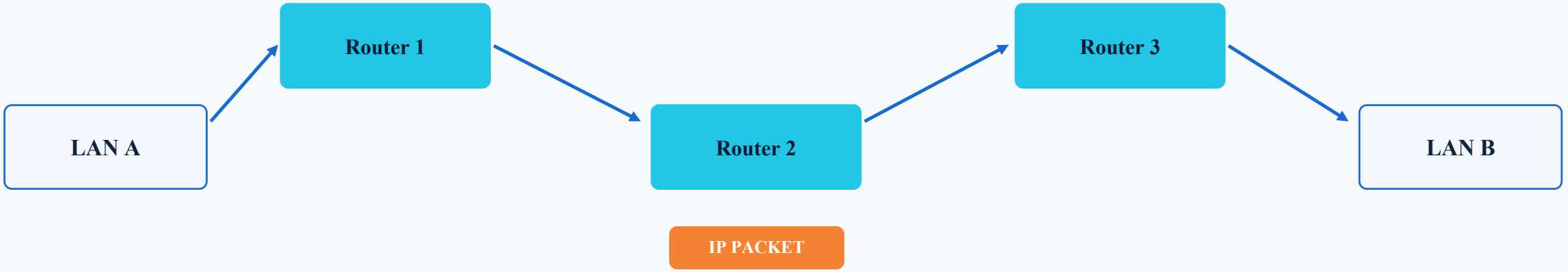
Feature	TP0	TP1	TP2	TP3	TP4
Segmentation	Yes	Yes	Yes	Yes	Yes
Error recovery	No	Yes	Yes	Yes	Yes
Multiplexing	No	No	Yes	Yes	Yes
Flow control	No	No	Yes	Yes	Yes
Retransmission	No	No	No	No	Yes
Reliable service	No	Yes	No	Yes	Yes

Teaching focus: recognize COTP as OSI transport material; do not let the TP0–TP4 table replace the TCP/UDP learning goals.

Network Layer

Packets cross different networks through routing



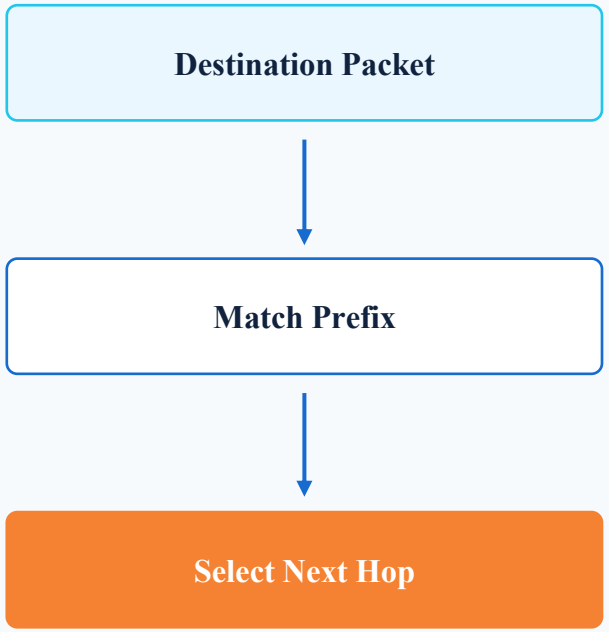


Key terms: logical addressing, packet, next hop and routing.

Routing Table Example

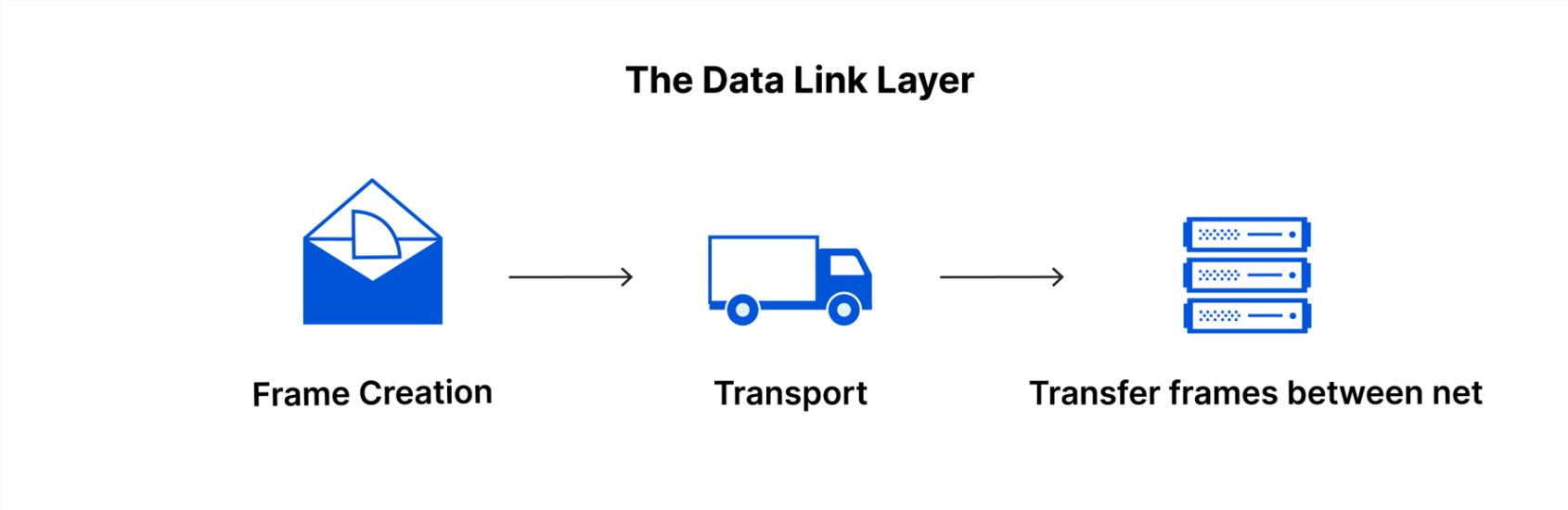
The destination address is matched to a route

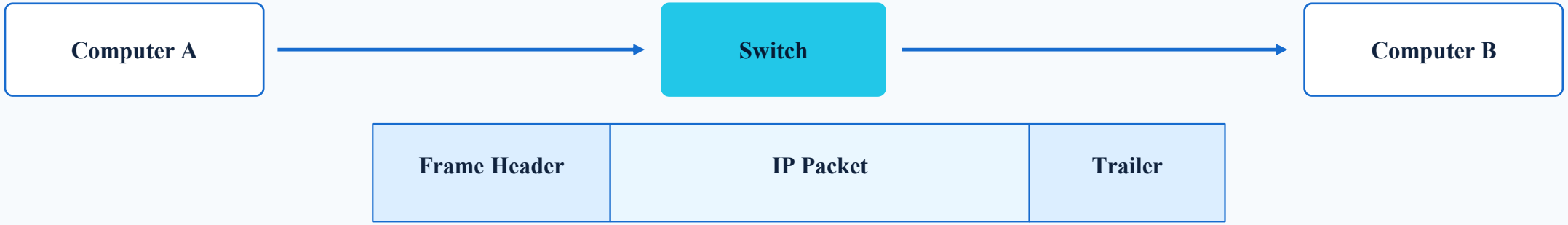
Destination	Prefix	Next Hop	Interface
192.168.1.0	/24	Direct	LAN
10.20.0.0	/16	10.0.0.1	G0/1
172.16.0.0	/12	10.0.0.2	G0/2
0.0.0.0	/0	ISP Gateway	WAN



Data Link Layer

Frames carry packets across one local link

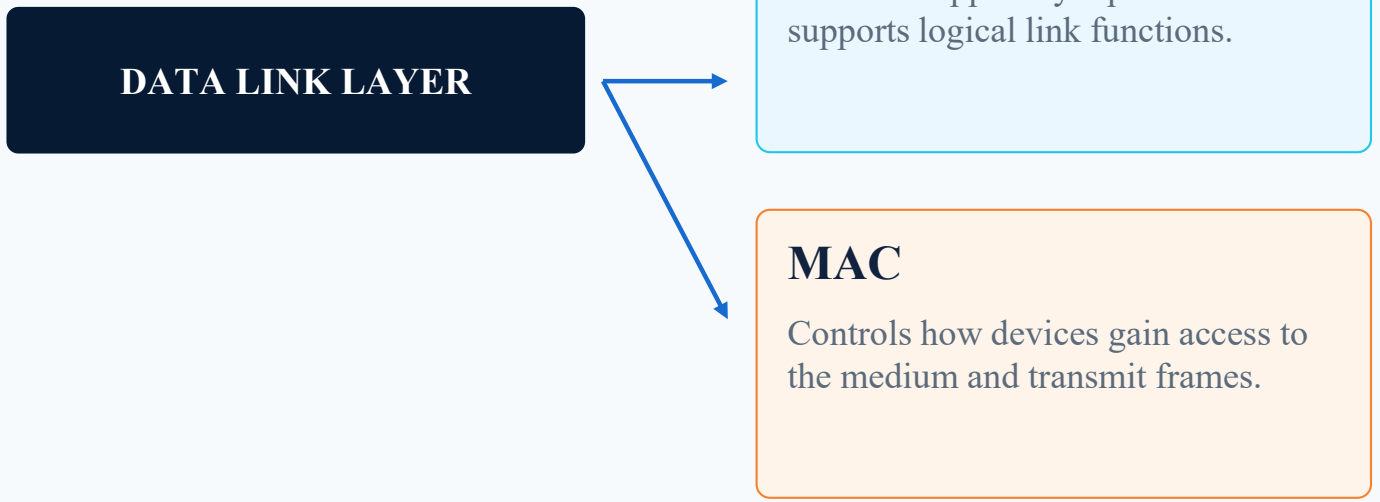




Network-layer packets can be wrapped in different data-link frames on different hops.

IEEE 802: LLC and MAC

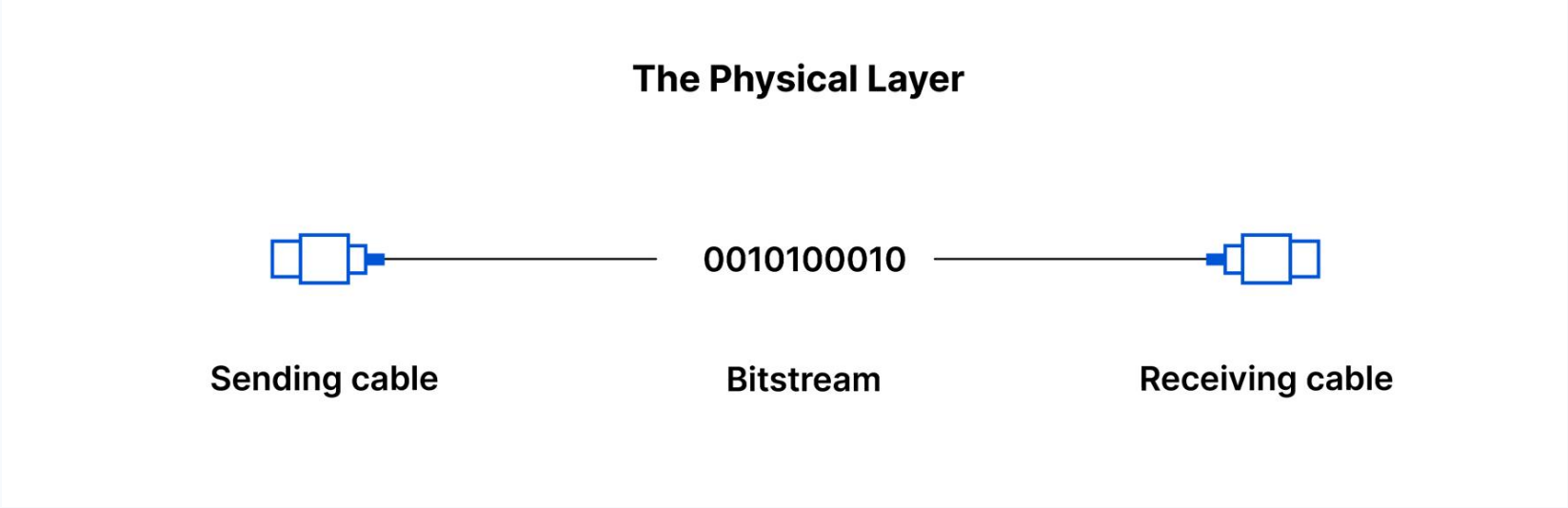
Data Link responsibilities are commonly divided into two sublayers



MAC sublayer ≠ only a MAC address. It is a media-access control function.

Physical Layer

Bits are represented as signals on real media



10110101

Electrical

Copper / Ethernet

Optical

Fiber

Radio

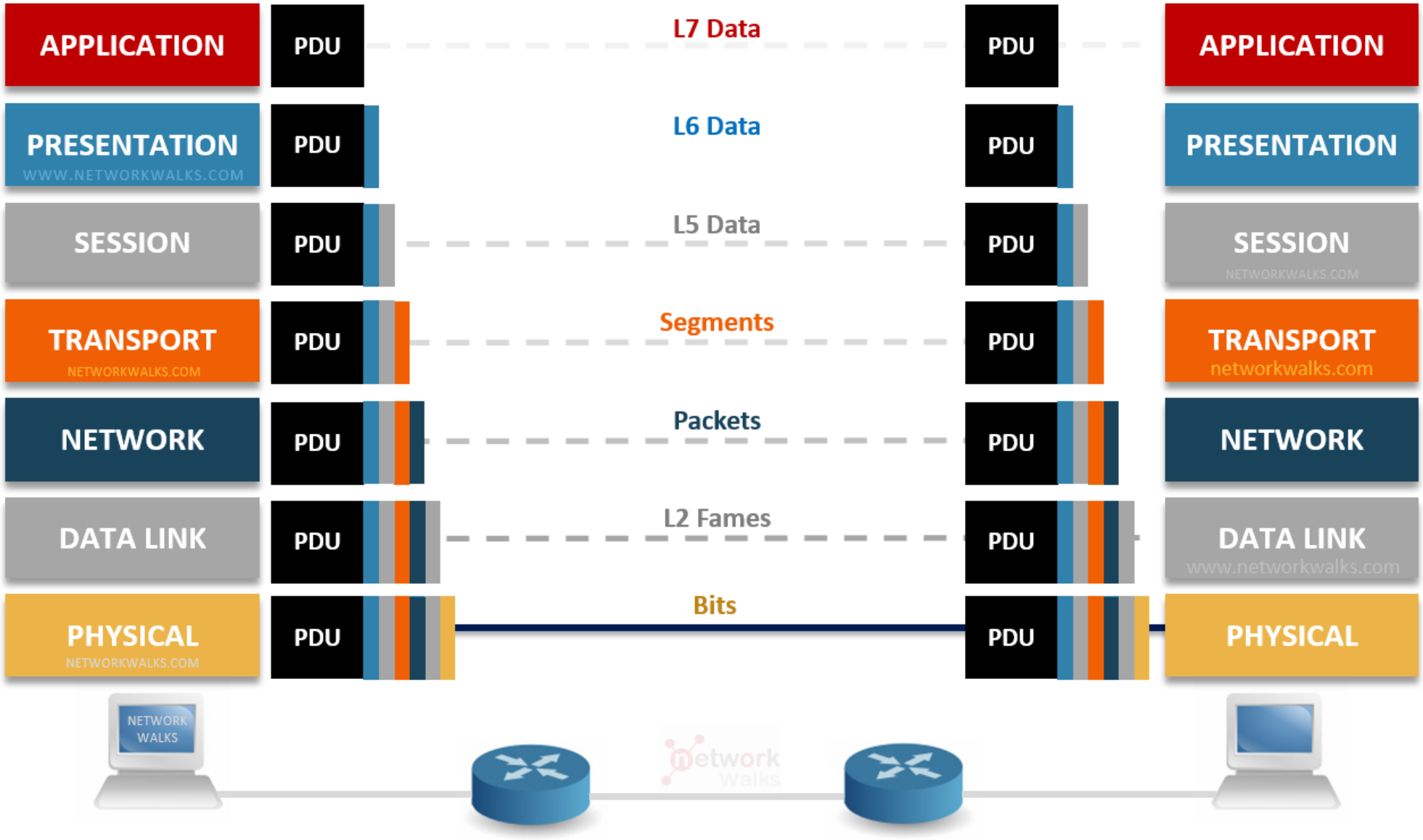
Wi-Fi / wireless

In Week 6, use physical media only as the bottom layer of communication. Detailed media types belong to Week 5.

OSI Model Summary

A review view for exams and troubleshooting

Layer	Main purpose	Data unit / concept	Example
Application	User/network service	Data	HTTP / SMTP
Presentation	Representation	Data	Formatting / encryption concept
Session	Session control	Data	Duplex / checkpoint
Transport	End-to-end delivery	Segment / Datagram	TCP / UDP
Network	Routing and addressing	Packet	IP
Data Link	Local delivery	Frame	Ethernet / MAC
Physical	Signal transmission	Bits	Cable / fiber / wireless


APPLICATION
PDU
L7 Data
PDU
APPLICATION
PRESENTATION
WWW.NETWORKWALKS.COM
PDU
L6 Data
PDU
PRESENTATION
SESSION
PDU
L5 Data
PDU
SESSION
NETWORKWALKS.COM
TRANSPORT
NETWORKWALKS.COM
PDU
Segments
PDU
TRANSPORT
networkwalks.com
NETWORK
PDU
Packets
PDU
NETWORK
DATA LINK
PDU
L2 Fames
PDU
DATA LINK
www.networkwalks.com
PHYSICAL
NETWORKWALKS.COM
PDU
Bits
PDU
PHYSICAL
NETWORK
WALKS
network
Walks

SECTION 3

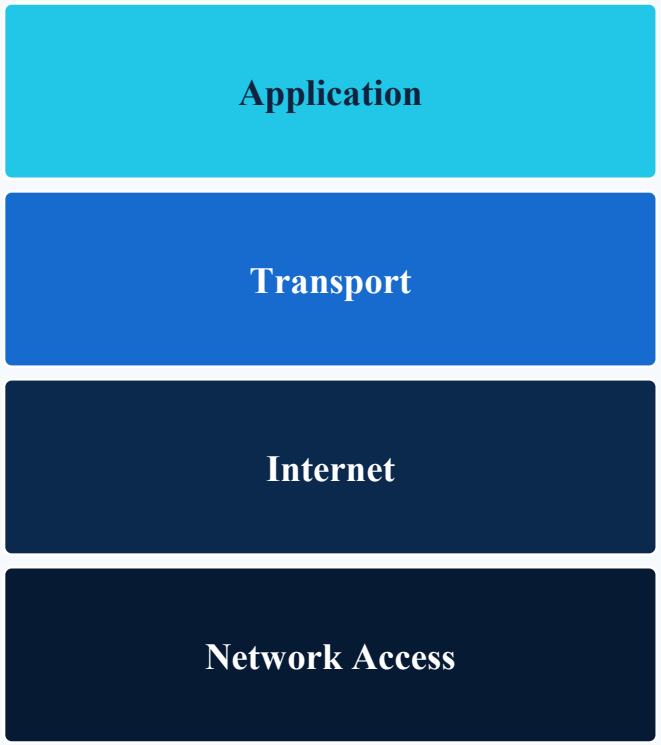
TCP/IP + IP Addressing

The practical Internet protocol architecture



TCP/IP Model

A practical architecture for Internet communication



Application

HTTP, DNS, SMTP and other application protocols

Transport

TCP or UDP between application processes

Internet

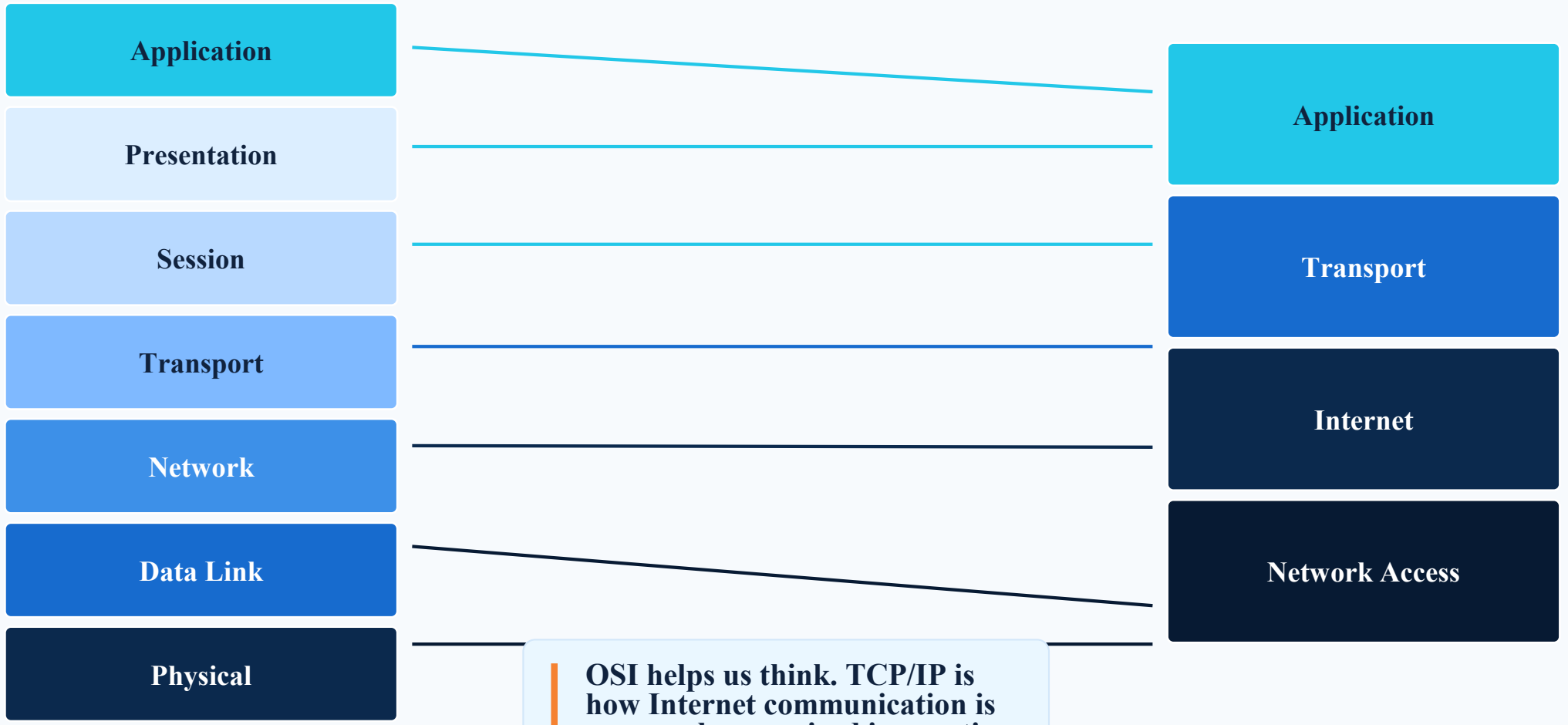
IP addressing and packet delivery

Network Access

Local network technology such as Ethernet or Wi-Fi

OSI vs TCP/IP Mapping

Related models with different purposes



OSI helps us think. TCP/IP is how Internet communication is commonly organized in practice.

Internet Protocol (IP)

Addressing and delivering packets across networks

What IP does

Delivers packets using source and destination addressing plus routing information.



IP is about packet delivery. It does not by itself guarantee reliable application communication.

IPv4

32-bit addresses written as four octets

192 . 168 . 1 . 10

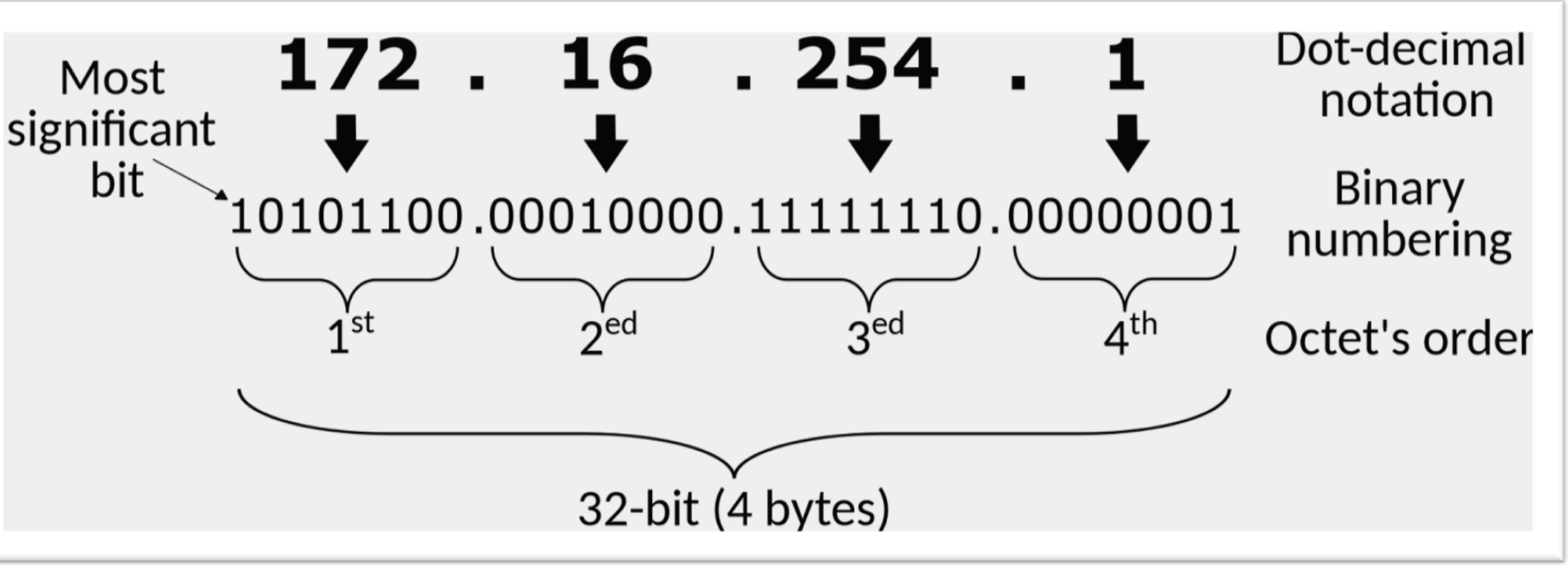


Address space

IPv4 uses 32-bit addresses.
2^32 = 4,294,967,296 possible values.

Common private IPv4 ranges

10.0.0.0/8
172.16.0.0/12
192.168.0.0/16


Most significant bit
172 . 16 . 254 . 1
Dot-decimal notation
10101100.00010000.11111110.00000001
Binary numbering
1st
2ed
3ed
4th
Octet's order
32-bit (4 bytes)

CIDR Notation

A compact way to write address + network prefix

192.168.1.0 / 24

IP address | **/24** | **Prefix length = network bits**



> **CIDR is the modern notation. Historical Class A/B/C ranges may appear in references, but CIDR is the key concept.**

Historical Classful Addressing

Useful context, but not the modern allocation model

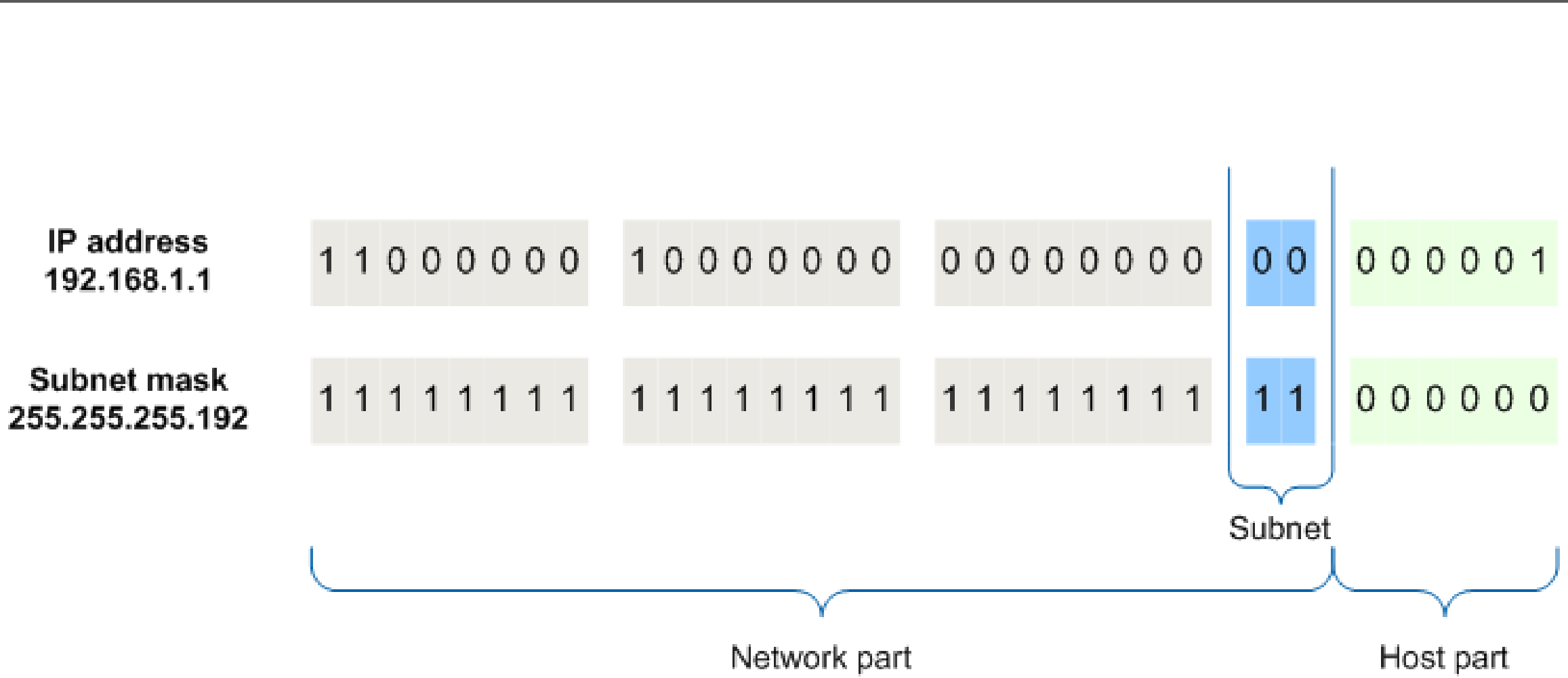
Class	Leading bits	Old prefix idea	Address range
A	0	/8	0.0.0.0–127.255.255.255
B	10	/16	128.0.0.0–191.255.255.255
C	110	/24	192.0.0.0–223.255.255.255
D	1110	Multicast	224.0.0.0–239.255.255.255
E	1111	Reserved	240.0.0.0–255.255.255.255

Do not confuse classful addressing with CIDR. CIDR uses prefix length such as /24.

Subnet Mask

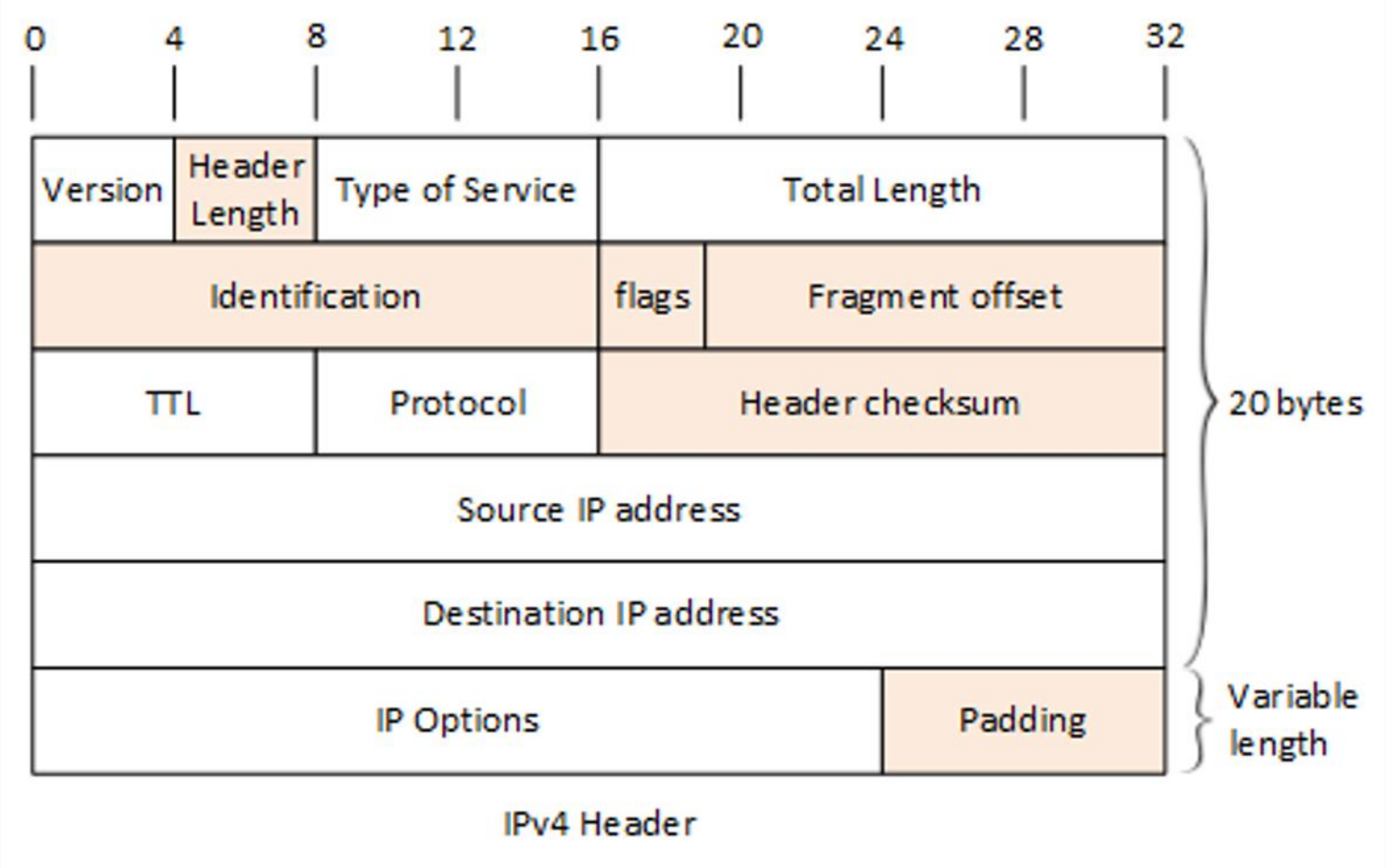
Another way to express the network and host parts

Item	Example	Interpretation
IP Address	192.168.10.25	Host address
Subnet Mask	255.255.255.0	Network mask
CIDR	/24	24 network bits
Network Part	192.168.10	Shared by hosts in this subnet
Host Part	.25	Specific host inside subnet


IP address
192.168.1.1
1 1 0 0 0 0 0 0
1 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0
0 0 0 0 0 1
Subnet mask
255.255.255.192
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1
1 1
0 0 0 0 0 0
Subnet
Network part
Host part

IPv4 Header

Important packet-control fields in a readable layout



Callouts

TTL limits packet lifetime.

Protocol identifies the next protocol, such as TCP or UDP.

Source/Destination guide delivery.

IPv6

128-bit addresses and a much larger address space

2001 : db8 : : 1

Why IPv6?

Larger address space for a world with many more connected devices.

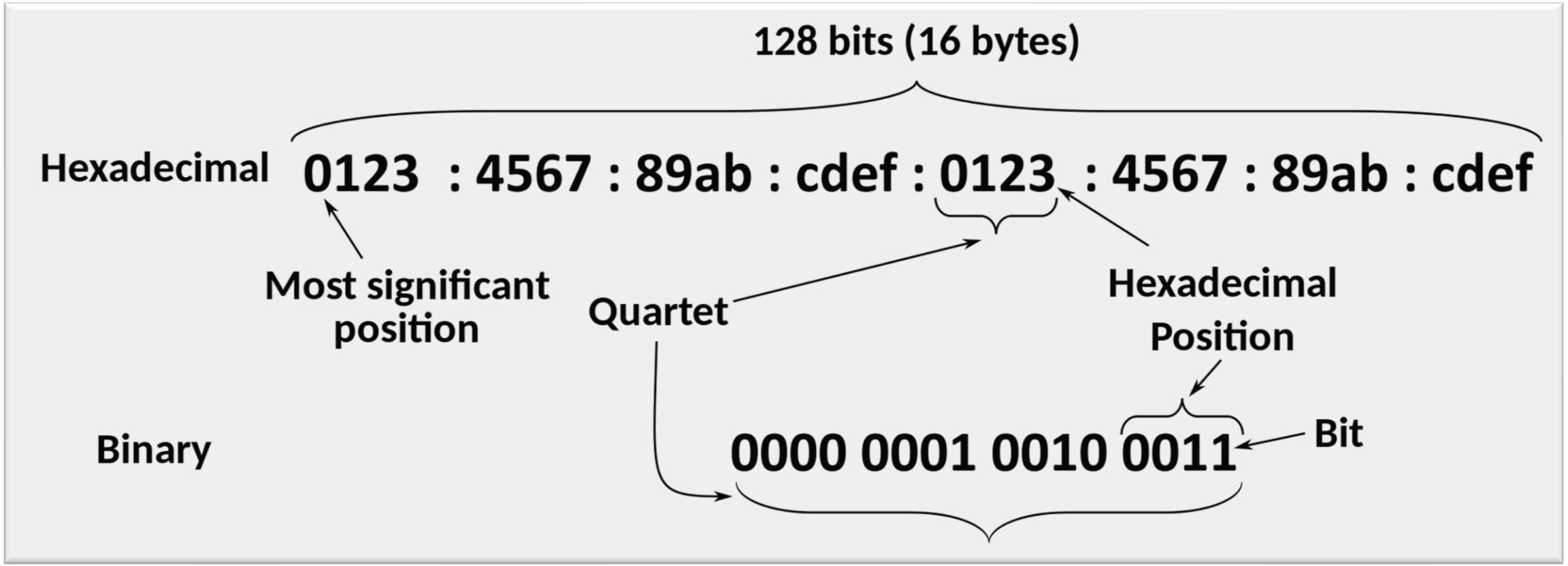
Address size

IPv6 uses 128-bit addresses.
2^128 ≈ 3.4 × 10^38.

Transmission types

Unicast
Anycast
Multicast

IPv6 is not just IPv4 with more digits; it also changes address format and several protocol behaviors.


128 bits (16 bytes)
Hexadecimal
0123 : 4567 : 89ab : cdef : 0123 : 4567 : 89ab : cdef
Most significant position
Quartet
Hexadecimal Position
Binary
0000 0001 0010 0011
Bit

IPv6 Header

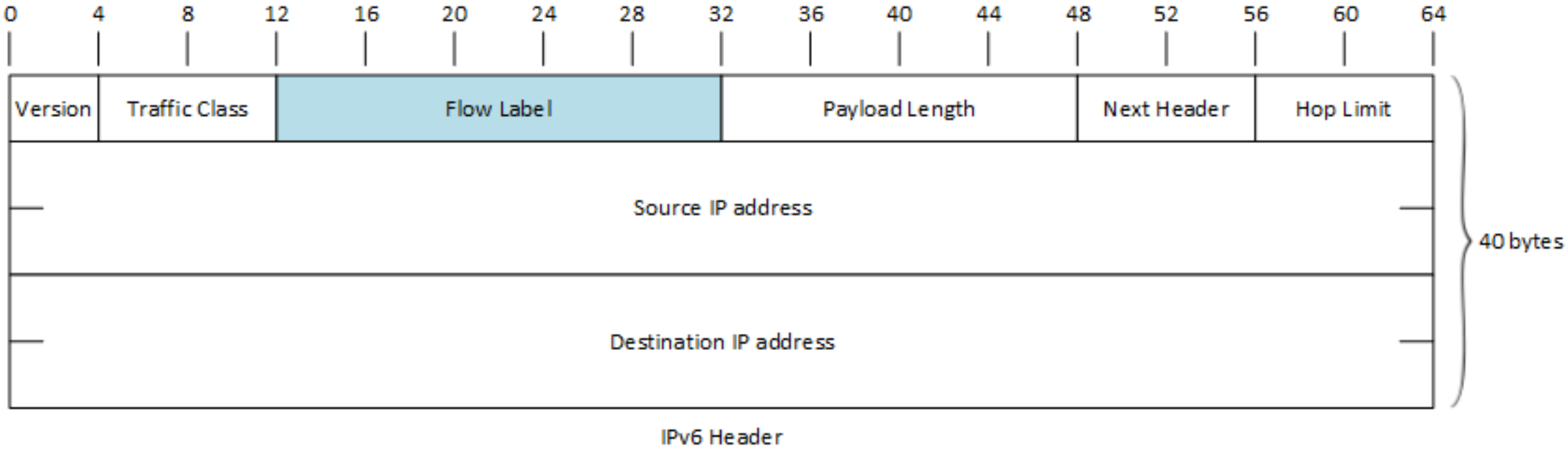
Simplified fixed base header with larger addresses

Field	Purpose
Version	Identifies IPv6
Traffic Class	Priority / traffic handling concept
Flow Label	Flow identification concept
Payload Length	Size of payload
Next Header	Identifies following header/protocol
Hop Limit	Similar purpose to TTL
Source Address	128-bit sender address
Destination Address	128-bit receiver address

IPv4 vs IPv6 header idea

IPv6 keeps a simpler fixed base header and moves optional information into extension headers.

The most visible change for beginners is the much larger address field.


0
4
8
12
16
20
24
28
32
36
40
44
48
52
56
60
64
Version
Traffic Class
Flow Label
Payload Length
Next Header
Hop Limit
Source IP address
Destination IP address
40 bytes

IPv6 Header

IPv4 vs IPv6

Key differences for introductory comparison

Aspect	IPv4	IPv6
Address size	32-bit	128-bit
Notation	Decimal octets	Hexadecimal groups
Private/local concept	10/8, 172.16/12, 192.168/16	ULA FC00::/7
Link-local	169.254.0.0/16	FE80::/10
Loopback	127.0.0.1	::1
Broadcast	Supported	No broadcast; uses multicast/other mechanisms

Embedded IPv4 Address Concepts

Compatibility representations between IPv4 and IPv6

IPv4-Compatible IPv6 Address

Historical / obsolete concept.
Example from source:
222.1.41.90 → ::de01:295a

IPv4-Mapped IPv6 Address

Used to represent an IPv4 address inside an IPv6 address structure.
Example:
222.1.41.90 → ::FFFF:222.1.41.90

> **Teaching note: identify legacy mechanisms clearly so students do not treat them as current best practice without context.**

SECTION 4

TCP & UDP

Two different approaches to transport communication



Transmission Control Protocol (TCP)

Connection-oriented communication with delivery mechanisms

Connection-oriented

A logical connection is established before data exchange.

Reliable mechanisms

Acknowledgements and retransmission support delivery.

Ordered stream

Data is reassembled in the intended order.

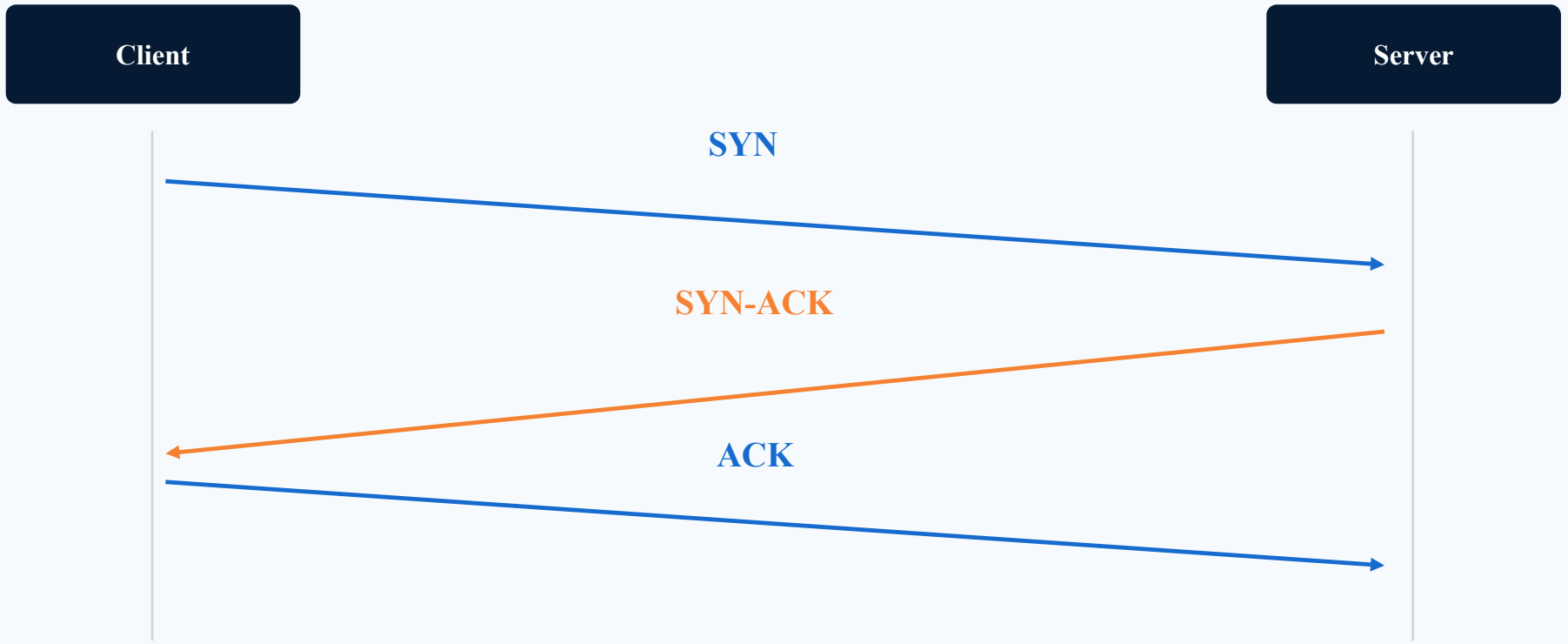
More overhead

Reliability mechanisms add control information.

> **Use TCP when the application needs completeness and order more than minimum delay.**

TCP Three-Way Handshake

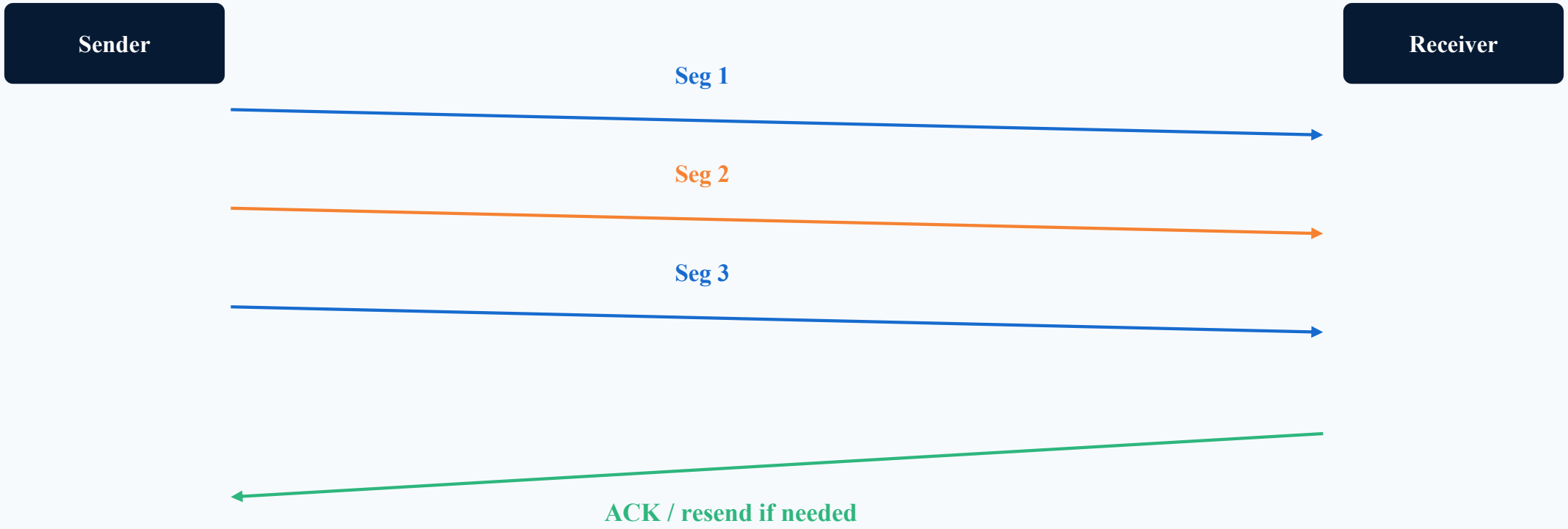
A simple sequence for establishing a TCP connection



The handshake coordinates both sides before normal data transfer begins.

TCP Communication Example

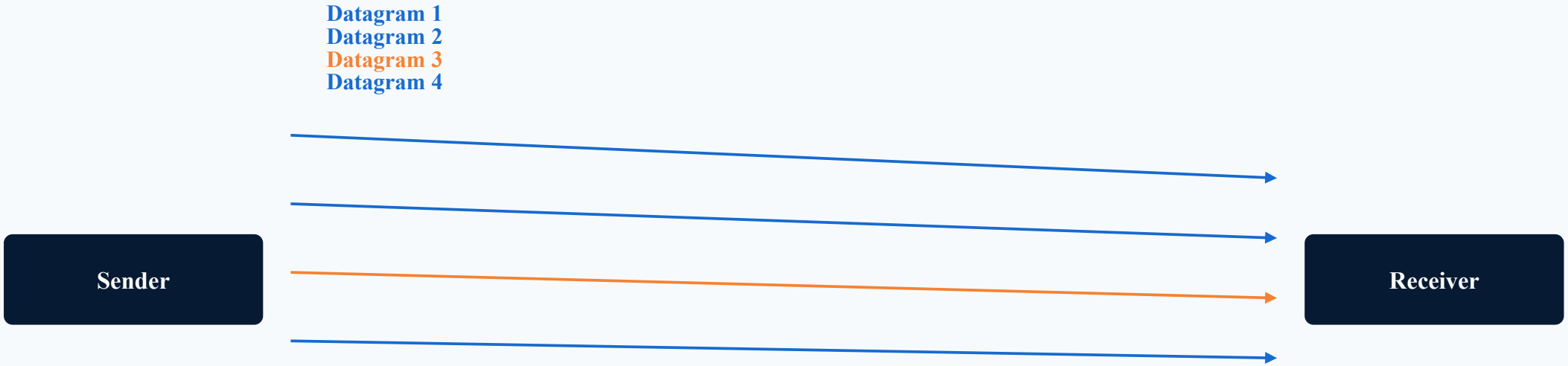
Sequence numbers and acknowledgements support reliable delivery



A missing segment is a useful classroom example of why reliability mechanisms matter.

User Datagram Protocol (UDP)

Connectionless datagram delivery without a handshake



No connection setup

Useful when timeliness matters or applications handle reliability themselves.

Examples

DNS queries
Real-time voice/video
Online games

Caution

Do not simply say “UDP is faster.”

TCP vs UDP

Reliability and timeliness are design trade-offs

TCP

Connection-oriented
Reliable delivery mechanisms
Ordered data stream
Acknowledgements
More transport overhead

Typical examples:
Web pages, file transfer, email

UDP

Connectionless
No built-in delivery guarantee
No built-in ordering
No handshake
Lower transport overhead

Typical examples:
DNS, voice/video, online games

Protocol Choice Examples

Different applications value different transport behavior

Scenario	Typical transport	Reason
Downloading a report	TCP	Every byte and correct order matter
Opening a web page	TCP	Reliable request/response is expected
Live video call	UDP concept	Late packets may be less useful than current packets
DNS lookup	UDP often	Small query/response with low overhead
Online game movement	UDP concept	Timeliness can be more important than perfect delivery

SECTION 5

ICMP

Control messages, error reporting and diagnostics



Internet Control Message Protocol (ICMP)

Network-layer control and diagnostic messages

ICMP role

Used by IP networks to report errors and support diagnostics such as reachability testing.



ICMP is not an application protocol like HTTP; it supports IP-layer control and diagnostics.

Important ICMP Message Types

A readable version of the detailed source table

Category	Meaning	Typical use
Echo Request / Reply	Reachability testing	ping
Destination Unreachable	The destination, host, protocol, or port cannot be reached	network error feedback
Redirect	Suggest a better route	routing control
Time Exceeded	Packet lifetime expired	traceroute concept
Parameter Problem	Header/options problem	error reporting
Timestamp / Reply	Time-related measurement concept	legacy/diagnostic context

Ping and Traceroute Concepts

Practical tools based on ICMP behavior

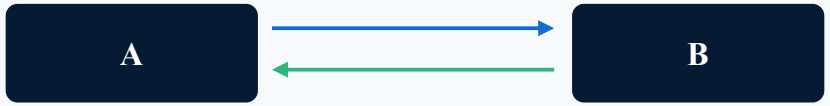
Ping

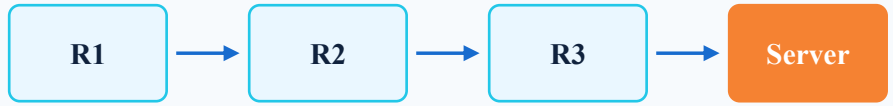
Echo Request → Echo Reply

Useful for checking whether a destination responds.

Traceroute

Uses Time Exceeded behavior to reveal hops toward a destination conceptually.





Troubleshooting by Layer

The OSI model becomes useful when something fails

Symptom	Likely layer to check	Example question
No link light / no Wi-Fi signal	Physical	Is the medium working?
Connected to LAN but local delivery fails	Data Link	Is MAC/local frame delivery working?
Cannot reach another network	Network	Is IP/routing correct?
Service reachable but app cannot connect	Transport	Is the port/protocol open?
Website or email app problem	Application	Is the service or request valid?

SECTION 6

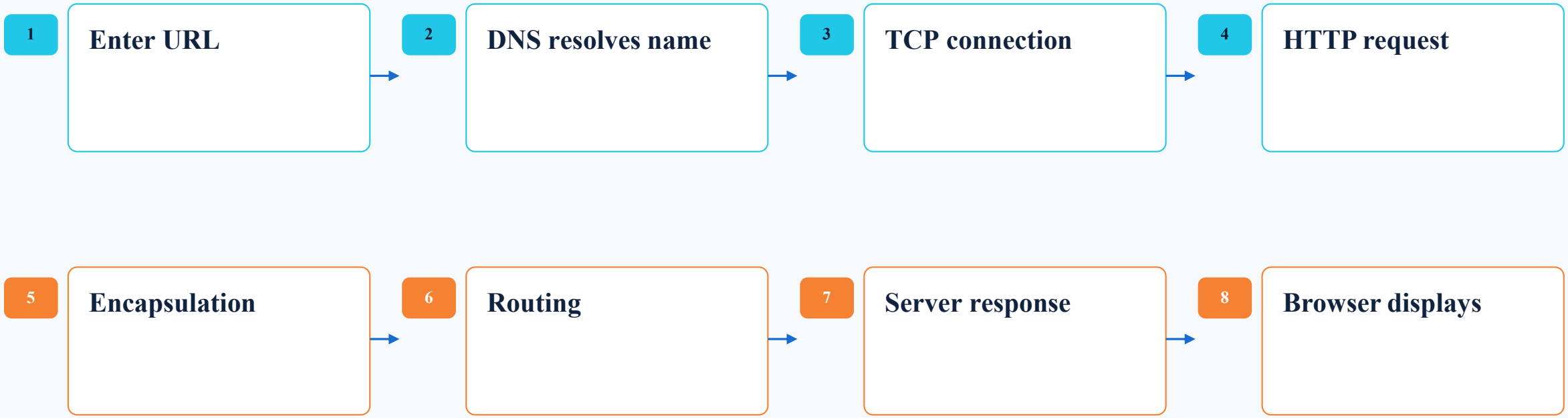
Applied Examples

From protocol concepts to real network situations



Example: Opening a Website

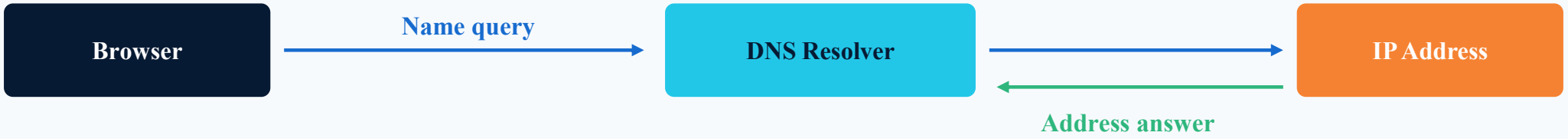
Putting OSI, TCP/IP, IP, TCP and DNS together



No single protocol does everything. Communication works because protocols cooperate across layers.

Example: DNS Name Resolution

Humans remember names; networks use addresses



DNS helps find an IP address. DNS does not carry the web page itself.

Example: MAC Address, IP Address and Port Number

Three identifiers answer three different questions

MAC Address

Which local network interface receives the frame on this link?

IP Address

Which host/network is the packet trying to reach?

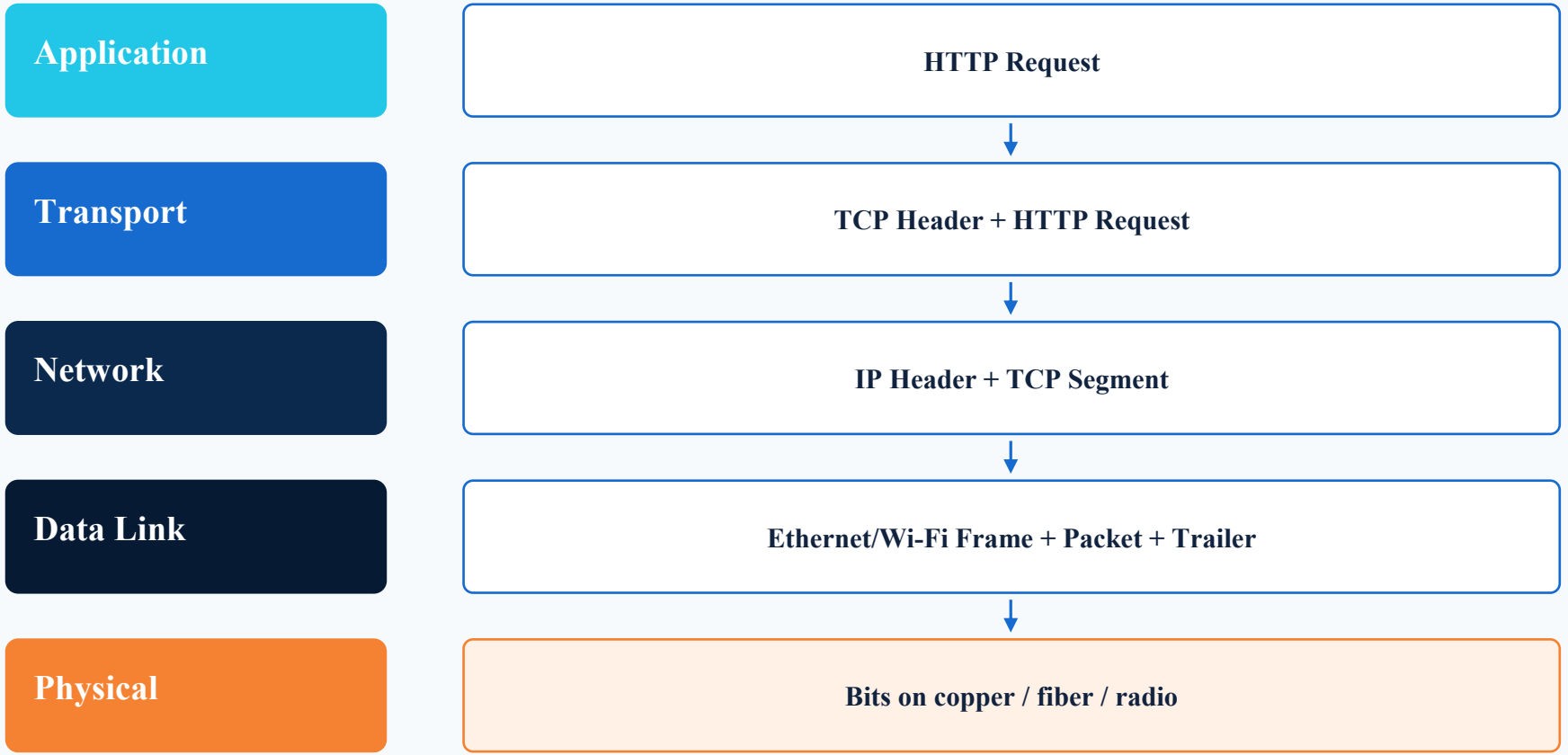
Port Number

Which application process should receive the data?

> **A port number is not the same thing as a physical router or switch port.**

Worked Example: Encapsulation

Opening a web page through layers



At the destination, decapsulation reverses this process.

Practice Example: CIDR and Subnet Mask

Read the prefix without advanced subnetting

Given

IP address: 192.168.10.25
Prefix: /24
Subnet mask: 255.255.255.0

Questions

1. Which part is the network?
2. Which part is the host?
3. What does /24 mean?

Expected idea

192.168.10 is the shared network part.
.25 identifies the host in that subnet.

Keep this activity conceptual; it prepares students for reading network information, not subnet engineering.

In-Class Activity: Build the Communication Stack

Pair activity, 5–7 minutes

Task

Place these labels in the correct order for opening a website:

HTTP
TCP
IP
Ethernet / Wi-Fi
Bits

Discussion

Which layer adds:

- port information?
- IP address?
- MAC/local link information?

Extension

Scenario:
A video call over Wi-Fi.

Which protocol behavior is useful?

Quick Review Quiz

Use before closing the lecture

1

Packet or frame: which one is used at the Data Link Layer?

2

Which protocol is connection-oriented: TCP or UDP?

3

What does CIDR /24 describe?

4

Does DNS carry the web page itself?

5

What ICMP message pair does ping use?

End-to-End Communication

Summary and questions

APPLICATION

What service?

TRANSPORT

TCP or UDP behavior + port

INTERNET

IP packet delivery

DATA LINK

Frame on local link

PHYSICAL

Bits as signals

ICMP

Control and diagnostics

Network communication works because multiple protocols cooperate across layers.

Thank You

Questions?

